

BAYESBALL IN THE OUTFIELD: QUANTIFYING DEFENSIVE AGGRESSION WITH A BAYESIAN HIERARCHICAL MODEL

SMT Data Challenge 2025

Team 15 - Undergraduate Division

Abstract

In this project, I came up with a metric that bridges the gap between current defensive metrics (e.g., OAA) and outfielder decision-making. This metric is called Aggression Over Expected (AOE), and it describes how much more aggressive an outfielder tends to be compared to the average player at their position. It is a cumulative metric, similar to how OAA is built, and takes into consideration a comparison between two probabilities: the probability that an outfielder *will* be aggressive, and the probability that the *average* player at the outfielder's primary position will be aggressive. These probabilities were then estimated using Bayesian Hierarchical Logistic Regression models. I also discuss how this metric is the first of its kind, its potential use cases, and even a leaderboard containing the top five and bottom five AOE scores.

Contents

1. Introduction.....	2
2. Data.....	2
3. Methodology.....	3
3.1 Quantifying Aggressiveness.....	3
3.2 Individual Player Probability.....	6
3.3 Average Positional Probability.....	7
3.4 Aggression Over Expected.....	7
3.5 Aggression Over Expected Limitations.....	8
4. Results.....	9
5. Discussion.....	11
6. Acknowledgements.....	13
7. References.....	13
Appendix A: Suggested Future Work.....	16
Appendix B: Models.....	18
Model 1: Catch Probability Model.....	18
Model 2: Average Jump Component Models.....	22
Model 3: Aggressive Probability Models.....	24
Model 3.1: Individual Player Model.....	24
Model 3.2: Position Average Model.....	28
Model 3.3: Model Specifications.....	30
Appendix C: Technical Details.....	32
Technical Detail 1: Launch Angle.....	32
Technical Detail 2: Exit Velocity.....	32
Technical Detail 3: Opportunity + Hang Time.....	34
Technical Detail 4: Distance Traveled + Needed.....	34
Technical Detail 5: Direction.....	34
Technical Detail 6: Sprint Speed.....	36
Technical Detail 7: Wall Proximity.....	36
Technical Detail 8: Jump.....	37
Technical Detail 9: Data Filtering.....	39
Appendix D: Description of the Data.....	40

1. Introduction

The diving catch, the run into the wall, and the last-second slide. These are all examples of aggressive defensive plays outfielders make that fire up crowds and end up on highlight reels somewhere. What these kinds of defensive plays also carry though, is risk. Missing a dive can turn a single into extra bases, or even an injury. As a result, are outfielders making the “*right*” decision to be aggressive, or are they making unnecessarily risky decisions that can end up hurting the team? Decisions like these clearly have an effect on the game, so how are they being accounted for?

When searching, it became apparent that there were very few, if any, widely adopted public approaches. One source I *did* find came from Sports Info Solutions (SIS). Mark Simon of SIS came out with an article about Carlos Correa's ability to make aggressive defensive plays more frequently than the rest of the league's shortstops in 2021¹. What's missing from his approach, however, is that there was no mention of a holistic metric that tells you something about Correa's aggressiveness. Current defensive metrics, (Outs Above Average (OAA), Defensive Runs Saved (DRS), Jump, and Fielding Run Value (FRV)), are all outcome-based, meaning they all measure what *has* happened. What these metrics fail to capture is the decision-making processes of the outfielders making the plays. This project aims to bridge the gap between outcomes and decision-making, but in particular, for risky, aggressive plays.

2. Data

The data for this project was provided by SportsMEDIA Technology and contains tracking information for hundreds of minor league games across two years. Each game is recorded at either 20 fps (every 50 ms) or 30 fps (every 33 ms), and provides 2D position data

¹ (Simon, 2022).

for each player and 3D position data for the ball ([Appendix D](#)). Given that this project focuses exclusively on outfield defense, I filtered the data to only include ball-in-plays (BIPs) that were first acquired by an outfielder. However, the data did come with its limitations. This includes missing or anomalous tracking points, and missing plays during games, leading to missing player IDs for those plays and fewer plays to work with overall ([Appendix C, Technical Detail 9](#)).

3. Methodology

This project uses a Bayesian framework to create a new defensive metric, Aggression Over Expected (AOE). Unlike today's defensive metrics that measure the success of outfielders, AOE measures how often outfielders *choose* to make aggressive plays. This distinction is essential because a player who is highly aggressive might not make the catch, so this metric serves best as a tool in identifying underlying behavioral patterns that *could* shape a successful defender. Heavily inspired by OAA, AOE is the cumulative sum of a comparison between two probabilities: the probability that the individual outfielder makes an aggressive attempt, and the probability of the average outfielder at their position making an aggressive attempt.

3.1 Quantifying Aggressiveness

The definition of aggression for this project is “when the outfielder puts their body on the line to make a play, e.g., dives, slides, or runs into the wall.” However, without access to a player's z-coordinate², aggressiveness had to be inferred based solely on two factors: their acceleration on their route and proximity to the outfield wall ([Appendix C, Technical Detail 7](#) for wall proximity calculation). To ensure that we focused on plays where outfielders had a legitimate reason to be aggressive, a BIP had to have at least a three star catch probability

² As a result, I could not account for aggressive leaps (e.g., robbing a homer).

difficulty level³ ([Appendix B, Model 3.1](#) for definition of a three star catch probability difficulty level). Plays with catch probabilities of zero, and a ratio of $\frac{\text{Distance Traveled by the player}}{\text{Distance Needed to make the catch}}$ below 0.95, were also excluded as these were clearly unreachable balls⁴.

To evaluate a player's aggressiveness as a measure of acceleration, I set a time window from 1.5 seconds before- to 0.5 seconds after- the landing point ([Appendix C, Technical Detail 3](#)). This interval was chosen to capture meaningful changes in velocity while avoiding noise from unrelated movements. The 1.5 seconds prior- effectively captures the rapid deceleration that occurs when a player prepares to dive⁵. The 0.5 seconds following- captures abrupt drops in velocity from wall collisions⁶.

For a play to be considered aggressive, a player's acceleration had to have dropped below -100 ft/sec^2 within the interval of 1.5 seconds prior to the landing point, or a player's acceleration had to have dropped below -200 ft/sec^2 from the 0.5 seconds following after- and be relatively close to the wall⁷. Regarding the 0.5 seconds following-, the player had to have *not* thrown the ball, as stopping to throw the ball to get it back to the infield quickly could drop acceleration in the same fashion as running into the wall.

Figure 1 below is an example play of a visible dive by the right fielder (RF), indicated by the sprint to the ball and staying still after the ball gets past, demonstrating my definition of aggressiveness.

³ This eliminated plays where a player slowed down to catch an easy fly ball, but it also meant that we implicitly omitted plays that could have been aggressive.

⁴ If a player traveled 90 feet and 93 feet was needed, they'd be within a few feet of the ball and it's perfectly reasonable that they could've made an aggressive attempt to get to that ball. This ratio would be around 0.97. It does not necessarily mean they covered 90 feet, but for simplicity's sake this will be our approach.

⁵ Velocity on a dive would decrease rapidly but not instantaneously.

⁶ Where an opposite force (the wall) is stopping the fielder's momentum.

⁷ The -100 ft/ms^2 and the -200 ft/ms^2 comes from investigating a number of plays and deeming these to be appropriate.

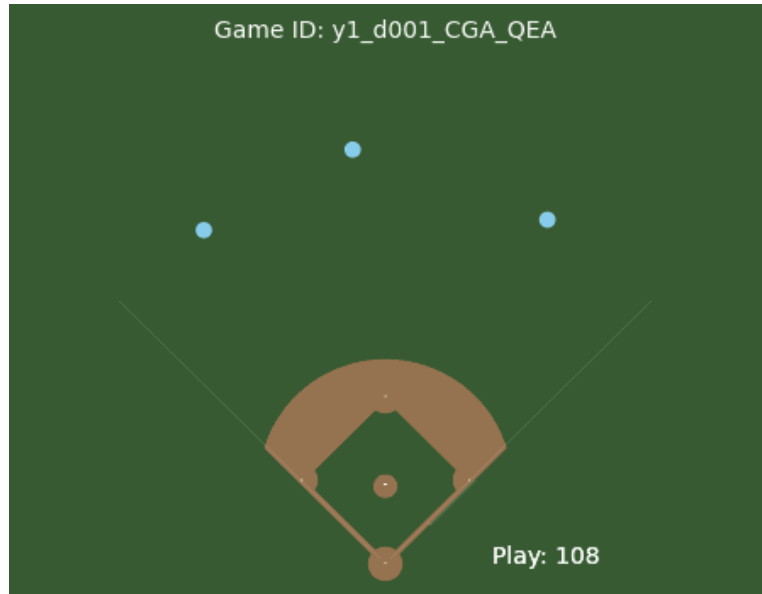


Figure 1: Play animation of a visible dive. Special thanks to Eddie Dew for the code used to generate this animation. If the animation is not playing, you can view it [here](#) (Right-click + “Open Link in New Tab”).

Figure 2 below is the RF’s velocity and acceleration from Figure 1 shown above. The marker for the dive can be seen as the sharp drop in acceleration on the right-side plot.

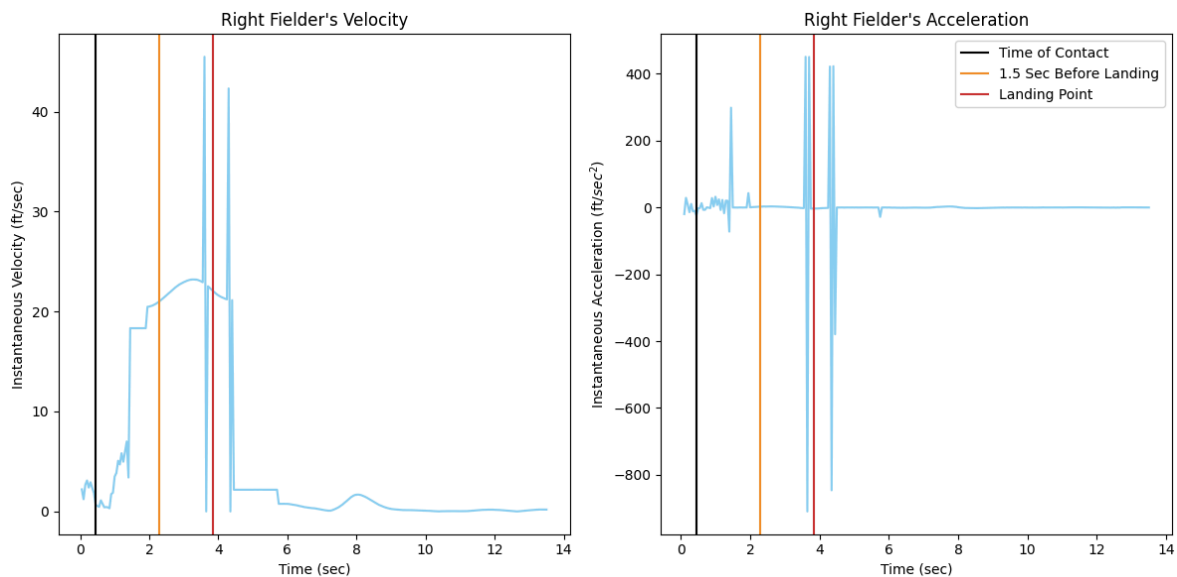


Figure 2: Velocity and acceleration line plots of the RF in Figure 2 above.

However, this approach is not without its limitations. The use of thresholds is inherently heuristic, and could be refined further with more time. Some *truly* aggressive plays may also go

undetected if they fall outside the specified intervals, or in the omitted, one-star to two-star, difficulty levels.

3.2 Individual Player Probability

Because individual outfielders have a widely varying number of opportunities in the data, often *very* small sample sizes, a Bayesian hierarchical logistic model was chosen. Bayesian hierarchical logistic models are a type of probabilistic model designed to estimate a binary variable, in our case, attempting an aggressive play or not, with the use of partial pooling⁸. The hierarchical structure of the model groups each player by their primary position, allowing us to estimate how aggressive each player tends to be, while simultaneously "borrowing strength" from their position group⁹. Player estimates are assumed to tend toward their positional average until they have a large enough sample size to stand on their own and retain more personalized estimates.

Because aggressive attempts are naturally uncommon, there will be considerable uncertainty when estimating these probabilities. The Bayesian aspect of the model directly addresses this concern by providing uncertainty intervals (similar to confidence intervals). Instead of providing a single probability, the model outputs a range of possible probabilities as a (posterior) distribution for each player ([Results](#), Figure 4). We'll use the (posterior) distribution's mean and call it $P_{\text{aggressive}, i, j, k}$, which is the probability that player i at position j attempts an aggressive play, given a BIP k and some context ([Appendix B, Model 3.1](#)).

⁸ Partial pooling treats each outfielder's observations as independent but related, and assumes that there is some underlying common distribution among each of the outfielders.

⁹ Borrowing strength allows information from the position group to contribute to the estimation of parameters of the player group.

3.3 Average Positional Probability

To establish a baseline expectation for the same given BIP and context, a hierarchical Bayesian logistic model was also used for the aggression probability of the average outfielder at a position. The difference between the two models is that this one removes the variability of player tendencies and focuses on the effect of position on aggressive attempts. We'll call this probability, $P_{\text{aggressive}, j, k}$, the probability the average player at position j attempts an aggressive play on the same BIP k and some context as the Individual Player Probability model ([Appendix B, Model 3.2](#)).

3.4 Aggression Over Expected

An AOE score is the difference between a player's individual aggression probability, $P_{\text{aggressive}, i, j, k}$, and their position's average aggression probability, $P_{\text{aggressive}, j, k}$, on a BIP k ¹⁰. Calculating this difference across all BIPs, for a given player, yields a single, holistic metric of that player's aggression relative to expectations, as illustrated in Figure 3 below:

$$\text{AOE}_{\text{total}} = \sum_{k=1}^N \text{AOE}_{i,j,k}$$

$$\text{AOE}_{i,j,k} = (P_{\text{aggressive}, i, j, k} - P_{\text{aggressive}, j, k})$$

Figure 3: Formula for AOE. Sum of (individual player's probability - average player probability at their position) on each BIP k for player i at their primary position j over N BIPs.

Essentially, AOE attempts to measure the aggressive nature of an outfielder, relative to positional expectations. A positive AOE indicates the player tends to play more aggressively, and

¹⁰ E.g., on some BIP, the CF Player A had a 20% probability of being aggressive, whereas the average CF only had a 12% probability of being aggressive. Player A would then have +0.08 points added to their AOE score, reflecting an 8% increase in aggressive probability over the expected probability.

a negative AOE indicates the player tends to play less aggressively; both with respect to their position's average.

And although AOE is a cumulative metric, its scores are not sensitive to small sample sizes, as the probabilities that go into AOE come from the models' (posterior) distributions. Players with small sample sizes will end up having probabilities that most resemble their position's averages, meaning their differences will hover around zero. AOE is also responsive to updates in tendencies for players as well, meaning, over time and with larger sample sizes, AOE can account for external factors¹¹ through Bayesian updating¹².

3.5 Aggression Over Expected Limitations

One limitation of AOE is that players will have higher scores solely due to having more opportunities. That's why AOE works best to compare players of the same position; this way centerfielders are not being compared to left fielders (LFs). But for AOE to be meaningful, a full season's worth of attempts would likely be required¹³.

A second limitation is that a player *will* be credited with an opportunity for every BIP towards the outfield. But in situations where nobody catches the ball because of miscommunication, it is not clear who deserves responsibility for the mishap. This means the closest fielder at the start of the play ends up receiving the responsibility for that mishap. In a situation where a center fielder (CF) calls off a LF (who was closer at the start of the play) but fails to make the catch, the "blame" would be assigned to the LF¹⁴.

¹¹ These being aging, change in defensive style, etc.

¹² Bayesian updating adjusts probabilities based on new evidence, so in our case, new evidence would increase a player's sample size, thereby helping dictate how much a player's aggressiveness deviates from their positional average.

¹³ As mentioned, due to the fact that aggressive plays are not common.

¹⁴ This is how OAA deals with these types of plays as well, as seen in this miscommunication between Kameron Misner and Chandler Simpson [here](#) (Right-Click + "Open Link in New Tab").

4. Results

Table 1 below is the AOE leaderboard alongside the player's OAA, and their successes and aggressive attempts by catch probability difficulty level¹⁵.

Top and Bottom 5 Outfielders by Aggression Over Expected (AOE)															
		5 Star Catch Prob (0-25%)					4 Star Catch Prob (26-50%)				3 Star Catch Prob (51-75%)				Total
Player	Position	AOE	OAA	Outs	Aggressive Attempts	%	Outs	Aggressive Attempts	%	Outs	Aggressive Attempts	%	Outs	Aggressive Attempts	%
Top 5															
RZQ-0274	CF	5.2	6.1	1	14	7.1	4	7	57.1	7	9	77.8	12	30	40.0
RZQ-0014	RF	2.8	0.9	0	1	0.0	0	3	0.0	5	6	83.3	5	10	50.0
RZQ-0409	LF	1.9	0.5	0	6	0.0	2	4	50.0	3	3	100.0	5	13	38.5
RZQ-0275	CF	1.7	-2.3	0	6	0.0	0	2	0.0	3	4	75.0	3	12	25.0
RZQ-0346	RF	1.4	5.6	1	5	20.0	2	2	100.0	6	6	100.0	9	13	69.2
Bottom 5															
YJD-0338	LF	-1.7	-0.1	0	5	0.0	1	1	100.0	2	2	100.0	3	8	37.5
YJD-0366	LF	-2.8	-0.3	0	3	0.0	0	1	0.0	4	4	100.0	4	8	50.0
YJD-0019	RF	-3.3	0.8	1	4	25.0	1	1	100.0	2	2	100.0	4	7	57.1
YJD-0327	CF	-3.9	0.5	0	4	0.0	0	2	0.0	3	3	100.0	3	9	33.3
YJD-0397	RF	-4.0	4.1	0	5	0.0	0	2	0.0	1	1	100.0	1	8	12.5

Table 1: AOE Leaderboard. Shows the top five highest and lowest AOE scores.

This leaderboard is insightful because it not only shows player aggressiveness, but also how successful they were at each difficulty level, inferring their approach on those kinds of BIPs. What's interesting is the disparity between RZQ-0274 and YJD-0327, in terms of aggression as CFs, so Figure 4 below is the comparison of both of these players' (posterior) distributions to the average CF.

¹⁵ Based on Statcast's catch probability leaderboard. (Statcast Catch Probability baseballsavant.com, 2025).

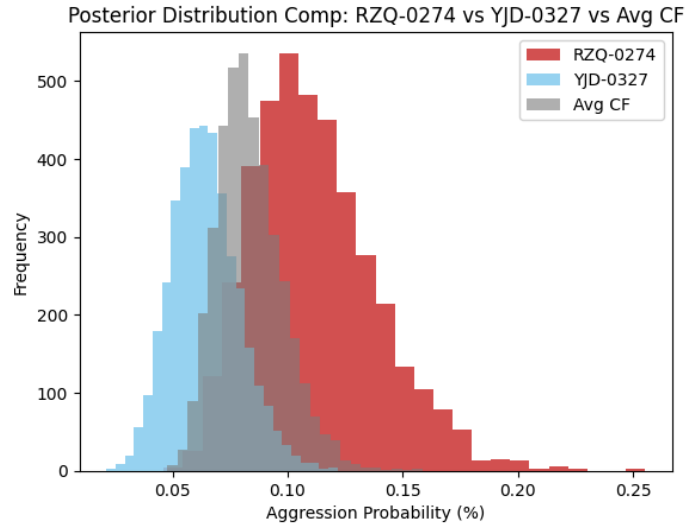


Figure 4: The posterior distributions of players RZQ-0274, YJD-0327, and the average CF. These distributions represent our updated belief about the player-specific and position-specific effects on aggressive probability, after observing the data.

The (posterior) distribution of each player and the average CF suggests that RZQ-0274 tends to be more aggressive than the average CF, and much more so than YJD-0327. The narrower the spread of the distribution, the more confident we are in that player's probability, which is evident in each player's case¹⁶.

Figure 5 below is a comparison between AOE and OAA for all outfielders who were on the field during a BIP (appearance) at least 100 times.

¹⁶ Although RZQ-0274's distribution spans a wider range, its concentration is not so spread out and is still between that 7-13% range, indicating that we still have high confidence in his aggressiveness being higher than average.

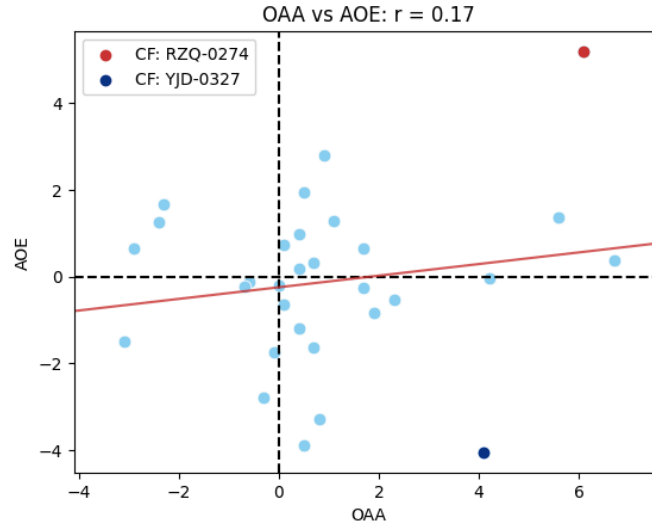


Figure 5: OAA vs AOE scatter plot. The red dot marks RZQ-0274, and the dark blue dot marks YJD-0327. The red line represents the line of best fit between the two metrics, with a weakly positive correlation ($r=0.17$).

We can interpret outfielders in quadrant I as ideal, quadrant II as inefficient, quadrant III as ineffective, and quadrant IV as conservative. RZQ-0274 and YJD-0327 represent opposite ends of the aggression spectrum, yet their similar OAAs suggest that both are effective defenders, just with differing defensive approaches.

5. Discussion

Given the behavioral nature of AOE, it serves best as a supplemental tool rather than as a standalone metric¹⁷. AOE would be best used in a player development or strategy setting, as, over a season, AOE could identify players who are overly-aggressive (or overly-*un*aggressive), relative to their peers, and discover the underlying defensive approach of players¹⁸. AOE could

¹⁷ AOE does not measure defensive skill, but rather the tendency to be aggressive, meaning a more aggressive player does not necessarily mean they're a better defender.

¹⁸ An application of AOE could be a comparison to a metric, such as OAA (given the similarities), in the off-season. E.g., a player with a high AOE but low OAA may be attempting too many low-probability plays and missing them, thereby needing development in refining their judgement and risk assessment. (Results, Figure 5).

also identify players who consistently exert above-average effort, regardless of whether the play results in an out^{19 20}.

AOE is the first step towards exploring the behavior and decision-making of outfielders and whether or not they *should* be aggressive and make a play. It helps provide context as to how outfielders are approaching BIPs, and it could even lead to research regarding mitigating injury risk. In 2021, Ronald Acuña Jr. made an aggressive leap towards the ball and tore his ACL in the process. When he came back from rehab (the second time), Acuña had this to say:

“I’d rather steal 30 and play the whole season as opposed to trying to steal 70, injuring myself and missing the whole year.”²¹

This same level of thinking could be applied to front offices regarding prioritizing the bats of their stars rather than their glove in the long-run. Teams would probably want their star to take it down a notch on defense once they get to a certain age and do not want to risk an injury²². AOE could potentially help teams identify *when* to tailor strategies surrounding a player, in order to create a balance between maximizing defensive output and health. I believe the use of incorporating Bayesian methods into the calculation of AOE is appropriate as it is able to account for the amount of uncertainty that comes with the different behaviors and decision-making of each player.

¹⁹ AOE is able to capture the non-outcome-based aspects of outfield defense.

²⁰ An example of this kind of player is RF Sal Frelick. Despite ranking 4th among MLB RFs in OAA with 6, his high-effort, often reckless defensive style is not fully captured in OAA or DRS, but *would* be with AOE. (Statcast Catch Probability baseballsavant.com, 2025).

²¹ (Gonzalez, 2025).

²² Teams have become *very* good at defensive positioning nowadays, and putting players in the right place could also help mitigate injury. AOE could identify who to emphasize when positioning.

6. Acknowledgements

I would like to thank SMT for organizing this challenge. I would like to thank Dr. Meredith Wills and Billy Fryer for their extensive assistance and advice throughout the entire process. I would like to thank Mark Simon, Davy Andrews, and Tom Tango for their help with answering my questions and being generous with their time and insights. I would also like to thank Gabe Riedel, for his function that organizes all of the data into a database, and Eddie Dew, for his animation function. Both of which were shared in the SMT Discord. Finally, I'd like to thank Chris Foncesbeck and Jim Albert for their insightful talks about Bayesian Statistics in baseball.

7. References

Andrews, D. (2022, October 5). Learning From Statcast's Outfield Jump Metrics. FanGraphs Baseball. <https://blogs.fangraphs.com/learning-from-statcasts-outfield-jump-metrics/>

Catch Probability | Glossary | MLB.com. (2025). MLB.com.

<https://www.mlb.com/glossary/statcast/catch-probability>

Foncesbeck. (2017). Hierarchical_Models_Sports_Analytics.ipynb [Jupyter Notebook]. GitHub.

https://github.com/foncesbeck/hierarchical_models_sports_analytics/blob/main/Hierarchical_Models_Sports_Analytics.ipynb

Gonzalez, A. (2025, March 14). Braves' Ronald Acuna Jr. has new plan after second ACL tear - ESPN. ESPN.com; ESPN.

https://www.espn.com/mlb/story/_/id/44235587/mlb-spring-training-2025-ronald-acuna-jr-atlanta-braves-rehab-acl

Hierarchical Models in Linear and Logistic Regression Basic Ideas. (n.d.).

<https://www.medicine.mcgill.ca/epidemiology/joseph/courses/EPIB-682/hier.pdf>

Jensen, S. T., Shirley, K. E., & Wyner, A. J. (2009). Bayesball: A Bayesian Hierarchical Model for Evaluating Fielding in Major League Baseball. *The Annals of Applied Statistics*, 3(2), 491–520. <http://www.jstor.org/stable/30244253>

Launch Angle (LA) | Glossary. MLB.com. (n.d.).

<https://www.mlb.com/glossary/statcast/launch-angle>

PyMC Labs. (2023, September 15). Developing Hierarchical Models for Sports Analytics with Chris Fonnesbeck. YouTube. <https://www.youtube.com/watch?v=Fa64ApS0qig>

Simon, M. (2022, January 5). Carlos Correa’s Amazing Slides, Dives, and Jumps. Sports Info Solutions.

<https://www.sportsinfosolutions.com/2022/01/05/carlos-correas-amazing-slides-dives-and-jumps/>

sklearn.linear_model.ElasticNet. (n.d.). Scikit-Learn.

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.ElasticNet.html

Spencer, S. (2024, September 22). Responses to XX Assessment Questions.

<https://www.linkedin.com/feed/update/urn:li:activity:7341911465357967363/>

Sprint Speed (SS) | Glossary. (n.d.). MLB.com.

<https://www.mlb.com/glossary/statcast/sprint-speed>

Statcast Catch Probability baseballsavant.com. (2025). Statcast Catch Probability.

Baseballsavant.com. https://baseballsavant.mlb.com/leaderboard/catch_probability

Statcast Directional Outs Above Average baseballsavant.com. (2025).

https://baseballsavant.mlb.com/leaderboard/outfield_directional_outs_above_average

What is Exit Velocity? – Baseball Connect. (2020). Baseball-Connect.com.

<https://baseball-connect.com/learn/what-is-exit-velocity/>

Zimmerman, J. (2014, February 6). All Fly Balls Are Not Created Equal. The Hardball Times.

<https://tht.fangraphs.com/all-fly-balls-are-not-created-equal/>

Appendix A: Suggested Future Work

There are a number of ways that I could've improved this project. One of the more important aspects that I could have improved is wall proximity ([Appendix C, Technical Detail 7](#)). More accurate estimations of the wall would have been helpful as it is incorporated into each of the models I used ([Appendix B](#)). The methodology that I came up with to estimate walls is not the greatest, and with detailed dimensions of the walls, those estimations would have been much more precise. Figure 6 below is an example of how my estimated walls could go with much better recalibrations and accurate dimensions.

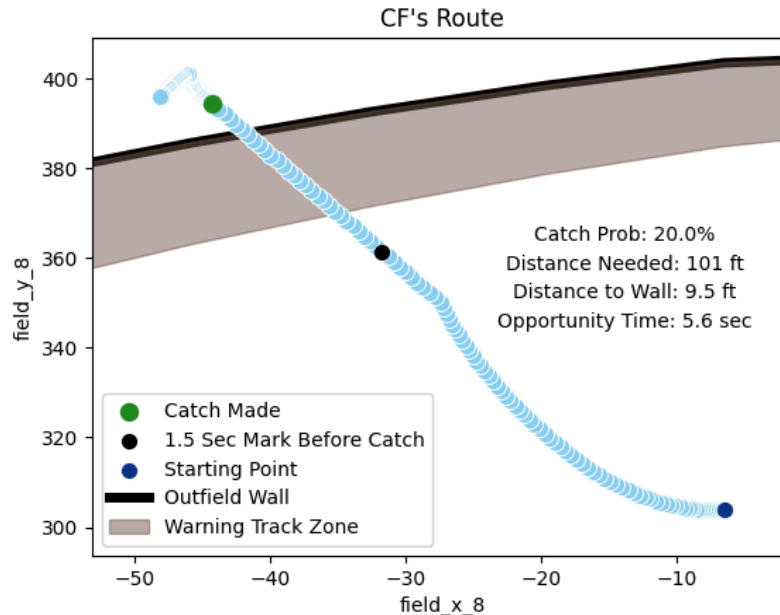


Figure 6: Visualizes the route the CF took on this play. He apparently makes the grab outside of the walls, yet the distance from the catch made point to the wall is 9.5 feet.

Similarly, another important aspect that would have been very helpful is having access to ball position data that had a z-axis based on the actual ground and not some arbitrary $z=0$ plane. With an accurate z-coordinate, I could have added an estimated height to the outfield's wall, but more importantly, I would have been able to include aggressive leaps as one of the aggressive plays I defined. Although, an accurate z-coordinate still would have limited me to only consider

leaps where the player had come down with the catch. A z-coordinate for a player's position would have also been very helpful, and would have made the labeling of aggressive plays much, much more accurate.

Another possible improvement would have been having access to more games across both years, as there were roughly only 275 total games played across both years. The issue with that is that these 275 games were separated into different series where it was difficult to have a big enough pool of players with insightful AOE scores. As a result, the 26 players, who had at least 100 appearances, all *only* played for the three main teams in the dataset: QEA, RZQ, and YJD. This makes comparison across players a lot less insightful than it could have been with more games. Not to mention, it would have allowed for a greater pool of players with enough sample sizes to help stabilize the Bayesian models even more. Having more games would have also allowed for a year-to-year comparison of aggressiveness, meaning I would have been able to incorporate year into the Bayesian models as a random effect and see the overall stickiness of AOE from year to year. This would have opened the door to exploring factors causing changes in aggressiveness (e.g., age, injuries, maybe an increase in confidence, etc.) and how predictable a player's aggressiveness is, both across two years. Having age for the players would have also been interesting, as it would have allowed for an investigation of whether or not a younger player actually did dive more than an older one. Most likely.

Another possible improvement would have been having access to a complete, clean version of *game_info* ([Appendix D](#)). This would have provided a complete set of game context (e.g., inning, number of outs, base runners, etc.), but also that there would be no missing player IDs. Meaning, each player would have received their proper credit, rather than the estimation I came up with to deal with the missing IDs ([Appendix C, Technical Detail 9](#)). Having proper

game context would have allowed me to investigate situational decision-making for players. Although, I do not think I would have incorporated it into the models as it most likely would have introduced a bit of noise. It would have still been interesting to see who was the most aggressive in each situation regardless.

An additional improvement includes refining player sprint speed, as I only accounted for a player's sprint speed based on defensive plays (that I had already filtered further). I would have rather have done it using baserunning instead, as these occurrences appear much more often than outfield plays.

Another additional improvement would have been features such as batted ball type (e.g., line drive, fly ball, pop-up), an encoded spray zone (e.g., shallow left, deep center, etc.), and bat handedness (lefty or righty). With enough data, these categories could have improved the Bayesian models by contextualizing the BIP further, but it is possible that it would have just introduced noise in the model's current state (due to the data it's trained on).

Finally, I would have liked to have explored taking a player's aggressiveness across all of their positions played into consideration, rather than just their primary position. Leading to questions such as: would a (primarily) corner outfielder become more aggressive by playing CF?

Appendix B: Models

Below are technical details of the models made for the project.

Model 1: Catch Probability Model

Given that the response variable of catch probability is binary (1 if the outfielder catches the ball, and 0 otherwise), a logistic regression model was the natural starting point. However, the

relationships between the features and our response variable, catching the ball, appeared non-linear and complex, which can be seen in Figure 7 below.

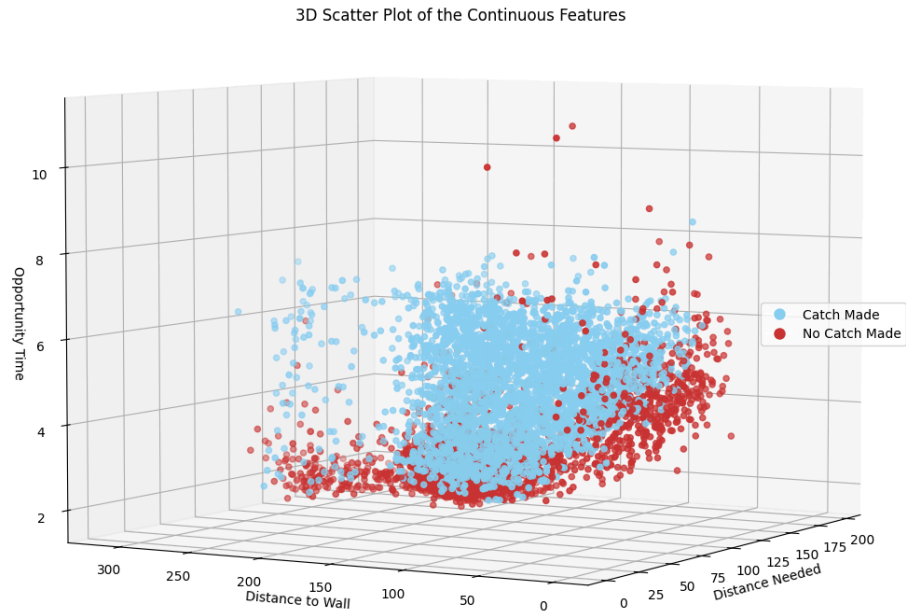


Figure 7: 3D scatter plot of the continuous features. The x-axis is Distance Needed, the y-axis is Distance to the Wall, and the z-axis is Opportunity Time. The hyperplane separating the classes, of whether a catch was made or not, is visibly non-linear.

As a result, a logistic generalized additive model (GAM) was chosen instead. A logistic GAM is useful here because it allows for non-linear relationships between each feature and the log-odds of the response, while maintaining additivity. This structure allows us to see how each feature directly contributes to catch probability, without the need to manually specify interaction terms (as standard logistic regression would have forced us to). No other models were considered due to a time constraint and the results of the GAM already being good.

The quantitative features were standardized to account for differences in scale (e.g., time vs distance), and direction was encoded as a categorical feature. A 75-25 train-test split with stratified sampling (maintains the proportion of successful catches) and pyGAM's grid search function was used to tune the model's optimal smoothing parameter (λ). This tuning was done

independently of cross-validation to avoid the computational burden of re-selecting λ within each fold. The model also included smooth terms for each continuous variable and a factor term for direction.

To validate the model, 10-fold stratified cross-validation with five repeats was used. Log-loss was used as the target metric because it measures the accuracy of predicted probabilities, something the other accuracy metrics do not do as effectively or at all. While the model's F1 score was reported, it was not chosen as the target metric, because in our case, false positives (predicted catch that is not made) and false negatives (predicted no catch but one is made) are not inherently detrimental.

The results of the model can be seen in Figures 8 and 9, and Table 2 below.

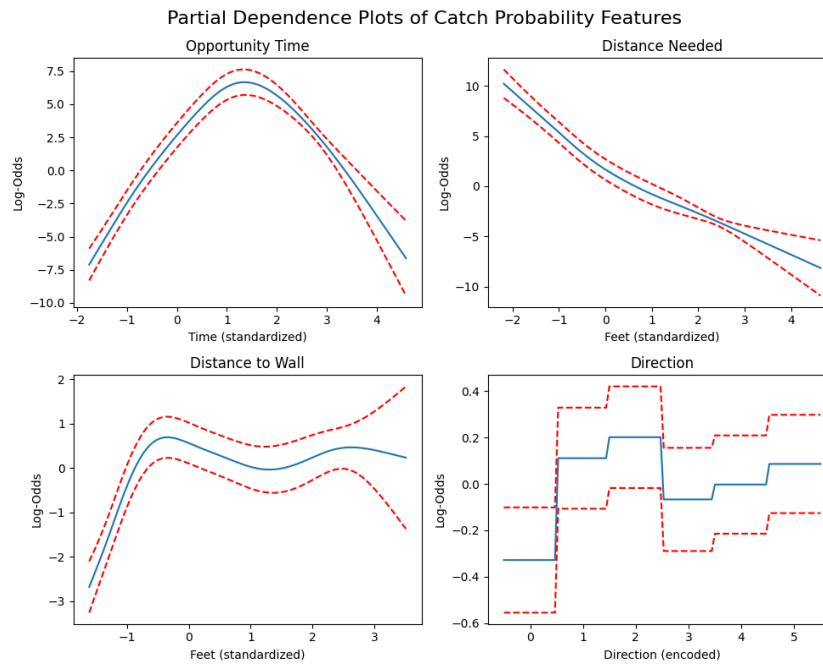


Figure 8: Partial dependence plot of each feature. The plots show the effect of each feature on the model's log-odds predictions where the blue line is that effect, and the red lines are that effect's 95% confidence interval.

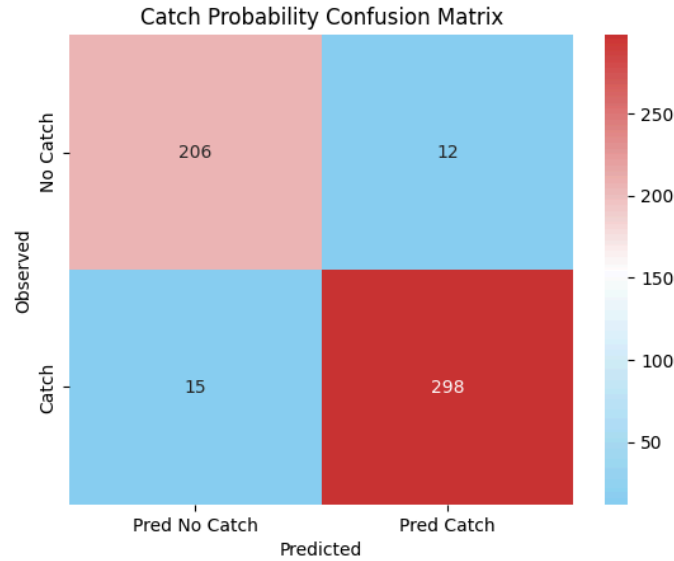


Figure 9: Confusion matrix for catch probability predictions. There are very few false negatives and positives, indicating a high accuracy on the test set.

Metric	Result
Log-loss	0.1624
F1 score	0.957

Table 2: Model Results. Shows the model's log-loss and F1 score. The F1 score is pretty good.

The results of the model was that it produced a log-loss of 0.1624, and an F1 score of 0.957. For comparison, if we were to naively say that the observed catch rate of outfielder across all BIPs is our catch probability “model,” our “model” would result in a log-loss of around 0.677 and an F1 score of 0.74. Meaning, we can confidently say that our catch probability logistic GAM produces pretty informative predictions.

To align with Statcast’s catch probability classifications, the raw predicted probabilities were rounded to the nearest multiple of five²³. The probabilities were then also categorized by difficulty (also to how Statcast does it), where a catch probability of 91-95% is categorized as

²³ (Catch Probability | Glossary | MLB.com, 2025).

one-star level difficulty, 76-90% as two-star, 51-75% as three-star, 26-50% as four-star, and 0-25% as five-star²⁴.

Model 2: Average Jump Component Models

Each component of Jump (Reaction, Burst, and Route) are compared relative to what an "average" player at that position would be expected to cover on the same BIP. This "average" is not calculated as a simple mean of that component, however, as BIPs are *not* created equally. Instead, the average is based on a complete pooling regression model. Because players at the different outfield positions will approach balls differently (e.g., CFs will tend to cover more ground than the corners), a separate model for each position and each component was used, resulting in nine total models. For simplicity, I used the same model for all nine cases.

Given that each component is a continuous variable, the model I chose to use was the Elastic Net linear model, as it combines both Lasso (L1) and Ridge (L2) regularization penalties, thereby simplifying the model and preventing overfitting. With a time constraint, I did not try other models because I wanted to prioritize regularization and the results of the Elastic Net models were good enough²⁵. The features used for each of these models include the same ones in the catch probability model ([Model 1](#)), with the addition of difficulty level (based on catch probability). The quantitative features were standardized to account for differences in scale (e.g., time vs distance), and direction and difficulty level were encoded as categorical features.

To validate each model, 10-fold cross validation with five repetitions was used, with a grid search over the Elastic Net's hyperparameters alpha α (controls the strength of the penalty),

²⁴ (Catch Probability | Glossary | MLB.com, 2025).

²⁵ The indicator if the results were good was if each model's RMSE was less than that component *and* position's standard deviation, meaning, the predictions are relatively close to the observed values. Each model did just that aside from the right fielder's burst and route models. With time constraints, this was acceptable.

and L1-ratio (balances the proportion of the penalty that comes from the L1 and L2 norms)²⁶.

The target metric for each model was RMSE (Root Mean Squared Error) as it is interpretable (the score remains in the original units of the response variable) and penalizes errors more than other metrics (e.g., Mean Absolute Error). The results of each model can be seen in Tables 3, 4, and 5 below.

Reaction

Position	Model Hyperparameters	RMSE
Left Field	$\alpha = 0.01$, L1-ratio = 0.1	1.738
Center Field	$\alpha = 0.01$, L1-ratio = 0.1	1.686
Right Field	$\alpha = 0.01$, L1-ratio = 0.1	1.875

Table 3: The results of the elastic net models for the Reaction component.

Burst

Position	Model Hyperparameters	RMSE
Left Field	$\alpha = 0.01$, L1-ratio = 0.1	5.911
Center Field	$\alpha = 0.01$, L1-ratio = 0.9	5.523
Right Field	$\alpha = 0.01$, L1-ratio = 0.9	5.865

Table 4: The results of the elastic net models for the Burst component.

Route

Position	Model Hyperparameters	RMSE
Left Field	$\alpha = 0.01$, L1-ratio = 0.9	7.741
Center Field	$\alpha = 0.01$, L1-ratio = 0.1	6.831
Right Field	$\alpha = 0.01$, L1-ratio = 0.1	7.683

Table 5: The results of the elastic net models for the Route component.

²⁶ (sklearn.linear_model.ElasticNet, n.d.).

Model 3: Aggressive Probability Models

Model 3.1: Individual Player Model

Given that our response variable is binary (1 if the player made an aggressive attempt, and 0 otherwise), a logistic regression model was once again appropriate. However, with the sheer amount of players in the dataset with small sample sizes, a logistic model would have made it difficult to estimate their *true* probabilities reliably. To account for this, I used a Bayesian Hierarchical Logistic Regression model, which enables partial pooling and a hierarchical structure of positions over players. As explained previously ([Methodology, Section 3.2](#)), the hierarchical structure allows each player to “borrow strength” from their position group. This way players with smaller sample sizes will have estimates closer to their positional average, and players who have enough of a sample size will stand on their own and retain more personalized estimates. No other classification models were considered for two reasons:

1. I wanted to estimate the aggressive probability for *each* player, thereby eliminating using any complete pooled model where we treat all players as collectively the same. However, introducing IDs into an ML model (to find the aggressive rate for each player) would lead to a *considerable* amount of overfitting, especially since a majority of the players in the data have small sample sizes, thereby eliminating a no pooled approach.
2. I *wanted* to explore using Bayesian methods and modeling in a baseball context.

The quantitative features were standardized to account for differences in scale (e.g., time vs distance), and direction and the closest teammate’s direction were encoded as categorical features. The model includes two levels of random effects: one for position and one for the individual player, where position-level effects are estimated first, and player-level effects are

centered around them. Meaning that players are nested within their primary position. Each random effect is also modeled using non-centered parametrization. This form of parametrizing allows Markov Chain Monte Carlo (MCMC) to sample efficiently, and reduces the likelihood of divergence²⁷. The model also included fixed effects on each feature's beta coefficient with a weakly informative prior ([Model 3.3](#)).

To approximate the posterior distribution of all model parameters, given the observed data, I used MCMC with PyMC's NUTS sampler, to generate 4000 posterior samples across all parameters (from four chains and 1000 samples per chain). Figure 10 below shows the energy plot that comes from MCMC²⁷.

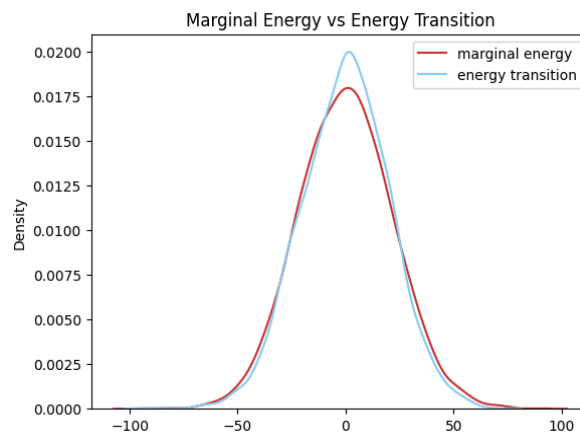


Figure 10: Energy distribution plot for the sampling of the Individual Player Model. The more similar the blue energy transition distribution is to the red marginal distribution, the more efficient the sampling is, resulting in near-independent samples from the posterior.

The results of the model can be seen in Figures 11 and 12 below.

²⁷ (Fonnesbeck, 2017).

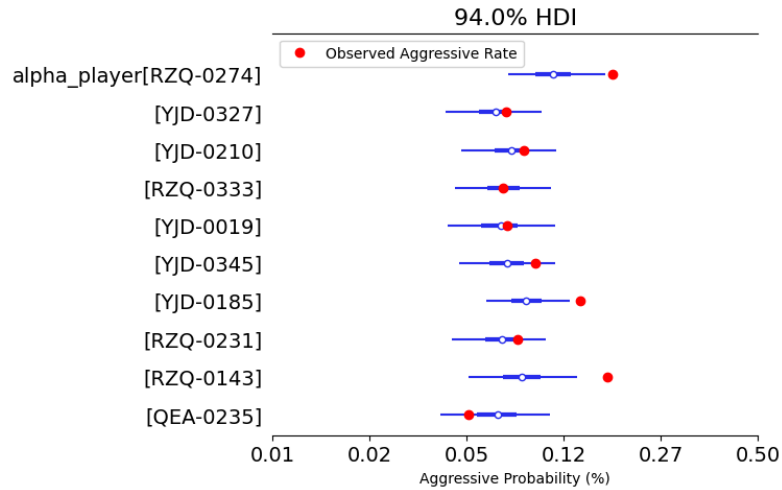
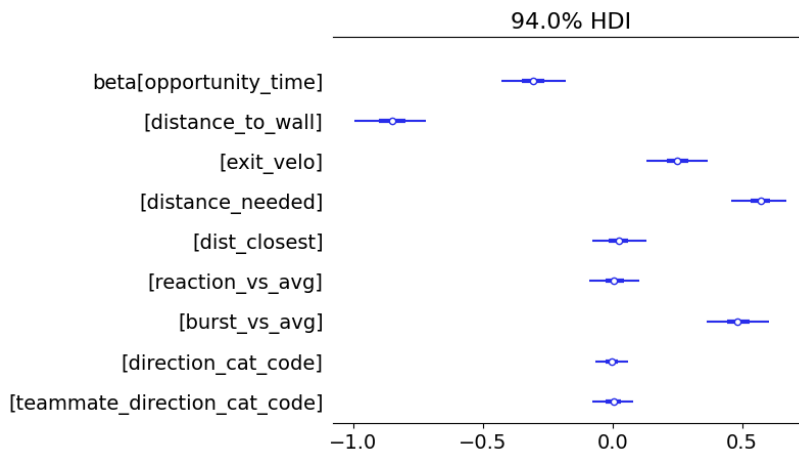


Figure 11: Forest plot of ten randomly sampled player's posterior estimates vs their observed aggressive rate. The purple lines represent each player's 94% highest density interval (HDI), similar to a confidence interval, and the red circle is that player's observed rate.

Figure 11 shows that the model is underestimating some players' observed aggressiveness²⁸. While the model adjusts for some context, it *cannot* fully approximate individual behavior tendencies because of the sample size issue. Especially aggressive players will be inherently underestimated if their *true* behavior lies in the tail of their posterior distribution because of partial pooling.



²⁸ The players shown were randomly sampled out from the group of players who had *at least* 25 total attempts made (they caught the ball or they were the closest fielder from the start of the play to when the ball landed).

Figure 12: Forest plot of posterior estimates for the model's coefficients. Purple lines show 94% HDIs, and the x-axis is in log-odds. Each coefficient reflects the average effect on aggressiveness from a one-unit increase in that feature, holding the other features constant.

To assess out-of-sample predictive accuracy, we used leave-one-out cross validation (LOO-CV), which estimates how well the model generalizes to unseen data without having to refit the model entirely²⁹. PyMC's LOO-CV has a built-in generalized Pareto distribution test²⁹, which estimated that each LOO-CV approximation in our model converged, meaning our model was well-calibrated.

One of the key limitations in our model is that while it produces posterior distributions for each player's aggressive probability, these estimates are still influenced by sample size. Because of partial pooling, players with the smaller sample sizes will have their posteriors shrink toward their positional group's averages, which is inherently under- or over-estimating a player's *true* tendencies³⁰. Given that this model produces a posterior distribution for each player's aggressiveness, it's used as the individual player aggressive probability aspect of AOE. A limitation of using this model to estimate each individual player's aggressive probability for AOE is that we're producing a distribution that considers uncertainty. The limitation comes in the fact that we *choose* to *only* use the posterior mean as the player's likelihood to be aggressive on that BIP. Thereby ignoring the uncertainty that comes with that player, however, for the sake of the simplicity of calculating the metric, I believe this is still a valid approach³¹.

²⁹ (Fonnesbeck, 2017).

³⁰ While this is a limitation of the model, it is certainly not a drawback. For players with smaller sample sizes, it's not possible to reliably estimate their *true* tendencies solely *because* of the small sample size. Shrinking their estimates toward their positional average is a reasonable approach.

³¹ Unlike OAA, which is built on hundreds of thousands of plays across the years, we had to come up with a way to deal with the rarity of aggressive plays, so I believe that this project comes up with a reasonable approach.

Model 3.2: Position Average Model

The positional average model is very similar to the individual player model ([Model 3.1](#)), with one big difference. While the individual player model takes each individual player into consideration through a random effect, this model completely removes that aspect and the variability that comes with the player random effect. The model instead focuses *solely* on the effect of position. The position average model is a Bayesian Hierarchical Logistic Regression model with a random effect *only* on position, modeled by non-centered parametrization³². The model *also* included fixed effects on each feature's beta coefficient with a weakly informative prior ([Model 3.3](#)).

This model also used MCMC with PyMC's NUTS sampler, to generate 4000 posterior samples across all parameters (from four chains and 1000 samples per chain). Figure 13 below shows the energy plot that comes from this model's MCMC results.

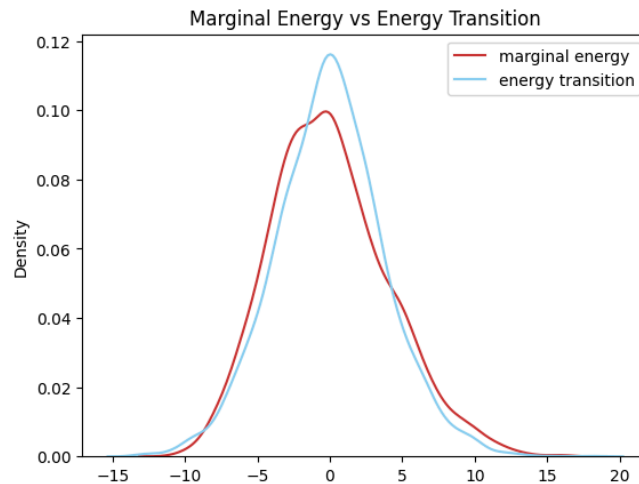


Figure 13: Energy distribution plot for the sampling of the Positional Average Model. The explanation for how this plot works can be found in Figure 10.

The results of this model can be seen in Figures 14 and 15 below.

³² For the same reason as mentioned in ([Model 3.1](#)).

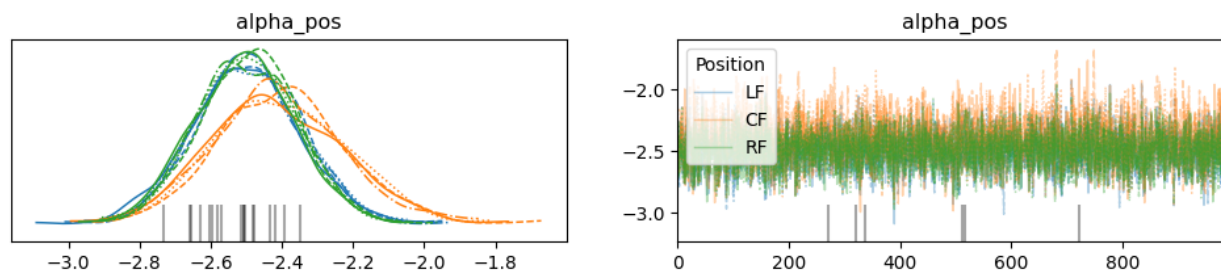


Figure 14: The posterior distributions of each position, where the x-axis is in log-odds and represents the aggressive probability.

Regarding Figure 14, the CF position seems to have more players with more aggressive tendencies, but the width indicates that there is less confidence in the CF's estimates compared to the corners positions with tighter distributions.

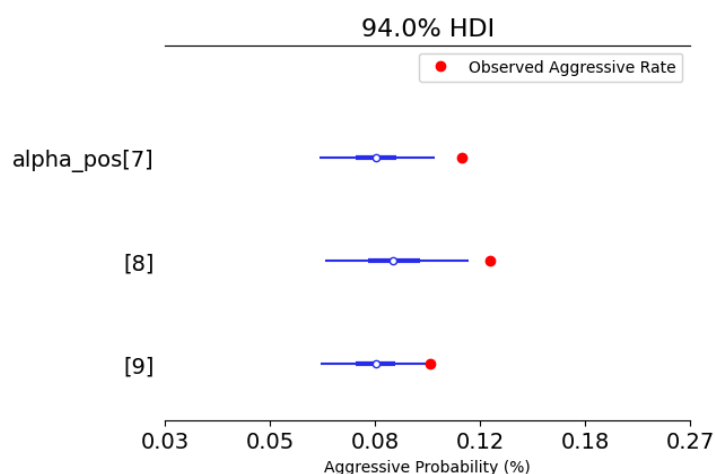


Figure 15: Forest plot of each position's posterior estimates vs that position's overall observed aggressive rate. The purple lines represent each position's HDI, and the red circle is that position's observed aggressive rate.

We can see in Figure 15 that *each* position's posterior is underestimating their observed rate. Because the positional average model omits player variability, which is absorbed in the observed rate, the partial pooling aspect is causing the conservative estimation of the posterior. One interesting thing to note is how LF has a slightly higher aggressive HDI_{97%}³³ than RF, which is not something you'd typically expect out of LFs (who are typically a team's *worst* defender).

³³ Is the equivalent of the 97.5th percentile in a confidence interval, but for Bayesian methods.

The 94% HDIs for the beta coefficients were also pretty much identical to that of the individual player model ([Model 3.1](#)).

The results from LOO-CV ([Model 3.1](#)) for this model was that each LOO-CV approximation in our model converged, meaning our model was well-calibrated. This model captures the broad positional trends and reinforces realistic trends (where CFs are still typically the most volatile and aggressive position), making this model appropriate within the calculation of AOE.

Model 3.3: Model Specifications

Specifications for the Individual Player Model can be seen in Figure 16 below:

$$\begin{aligned}
 z_{\alpha_j} &\sim \mathcal{N}(0, 1) \\
 \sigma_{\alpha_j} &\sim \text{HalfNormal}(1) \\
 \mu_{\alpha_j} &\sim \mathcal{N}(0, 1) \\
 \alpha_j &= \mu_{\alpha_j} + \sigma_{\alpha_j} \cdot z_{\alpha_j} \\
 z_{\alpha_i} &\sim \mathcal{N}(0, 1) \\
 \sigma_{\alpha_i} &\sim \text{HalfNormal}(1) \\
 \alpha_{i[j]} &= \alpha_j + \sigma_{\alpha_i} \cdot z_{\alpha_i} \\
 \beta &\sim \mathcal{N}(0, 1) \\
 y &\sim \text{Bernoulli}(\text{logit}^{-1}(\alpha_{i[j]} + \beta \cdot \mathbf{X}^\top)) \\
 y_k &\sim \text{Bernoulli}(\text{logit}^{-1}(\alpha_{i[j],k} + \beta \cdot \mathbf{X}_k^\top))
 \end{aligned}$$

Figure 16: Model specification of the individual player model. Where i represents the player effect, j the position effect, and k the BIP.

The choice to use the Bernoulli comes from the fact that making an aggressive attempt or not is a Bernoulli realization and is reinforced by the use of it in Jensen's model³⁴, and it models

³⁴ (Jensen, Shirley, & Wyner, 2009).

our logistic regression model³⁵. A HalfNormal(1) weakly-informative prior is used for each random effect's standard deviation, following Fonnesebeck's Bayesian hierarchical model implementation³⁶. The non-centered parametrization can be seen as the parameter z , which are just independent standard normals, however, they are used to scale the μ parameter of each random effect³⁴. The rest of the priors in the model are weakly-informative independent standard normals due to a limited domain knowledge within player aggressiveness³⁷.

Specifications for the Position Average Model can be seen in Figure 17 below:

$$\begin{aligned}
 z_{\alpha_j} &\sim \mathcal{N}(0, 1) \\
 \sigma_{\alpha_j} &\sim \text{HalfNormal}(1) \\
 \mu_{\alpha_j} &\sim \mathcal{N}(0, 1) \\
 \alpha_j &= \mu_{\alpha_j} + \sigma_{\alpha_j} \cdot z_{\alpha_j} \\
 \beta &\sim \mathcal{N}(0, 0.5) \\
 y &\sim \text{Bernoulli}(\text{logit}^{-1}(\alpha_j + \beta \cdot \mathbf{X}^\top)) \\
 y_k &\sim \text{Bernoulli}(\text{logit}^{-1}(\alpha_{j,k} + \beta \cdot \mathbf{X}_k^\top))
 \end{aligned}$$

Figure 17: Model specification of the position average model. Where i represents the player effect, j the position effect, and k the BIP.

The specifications in the Position Average Model are pretty much the same, with the key differences being the use of position as the *sole* random effect and a more regularizing Normal distribution on the beta coefficients. A standard deviation of 0.5 was chosen in the coefficient's prior, instead of 1, to make up for the variance lost by omitting the player effect, and keep the model from overfitting.

³⁵ The inverse logit function *becomes* logistic regression.

³⁶ (Fonnesebeck, 2017).

³⁷ Even though an informative prior could've helped decrease the uncertainty in the model's posterior estimates by a considerable amount, with a lack of domain knowledge, I did not want to introduce any bias or straight *wrong* predictions.

Appendix C: Technical Details

Below are technical details of some calculations and formulas I used throughout the process.

Technical Detail 1: Launch Angle

Launch angle was calculated using the formula: $\theta = \tan^{-1}\left(\frac{\Delta \text{Vertical Position}}{\Delta \text{Horizontal Position}}\right)$ after 0.1 seconds (or 100 ms) before gravity had *too* much of an effect on the ball's trajectory, and can be visualized in Figure 18 below. Credit to Kique Ruiz in the Discord for the exact formula.

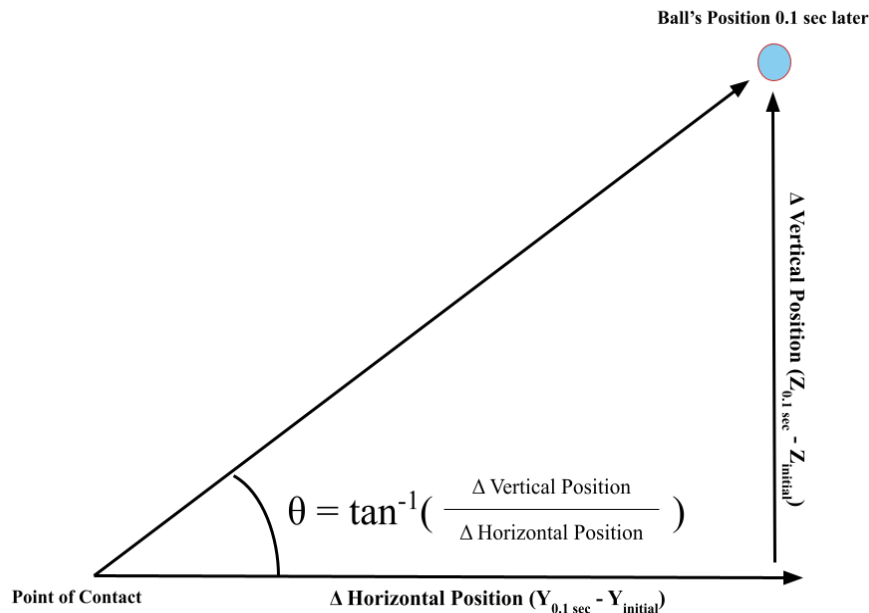


Figure 18: Visual of the geometry behind launch angle, including the point of contact, the vectors of change in horizontal & vertical position, the ball's straight line trajectory, and the ball's position after 0.1 sec.

Technical Detail 2: Exit Velocity

Exit velocity was calculated by taking the median speed of the ball 0.4 to 0.6 seconds after contact, and capping it at 120 mph³⁸. Speed is calculated by taking the magnitude of the velocity vector between consecutive timestamps. Initially, exit velocity was calculated by taking the

³⁸ The hardest hit ball in the minors, that *was not* hit by Oneil Cruz as he had already made his debut at this point, was registered at 119.6 mph. (Used Statcast's Minor League Search for this).

speed of the ball after 2 frames, however, there were too many extreme cases of exit velocities (e.g., one player had an exit velocity north of 200 mph). Taking the speed after 0.4 to 0.6 seconds meant the ball lost some of its velocity due to drag and gravity, however, this made the values more realistic. The distribution, as a result, led to the average of the exit velocities being approximately 87% of the average of 89 mph in the majors³⁹, so I scaled the exit velocities by about 1.15 to fit the MLB averages. You can see this change in exit velocities in Figure 19 below. The overall shape of the distributions has not really changed much, as seen by the kernel density estimation plots overlaying the histograms, besides there being more batted balls that were hit 110-120 mph.

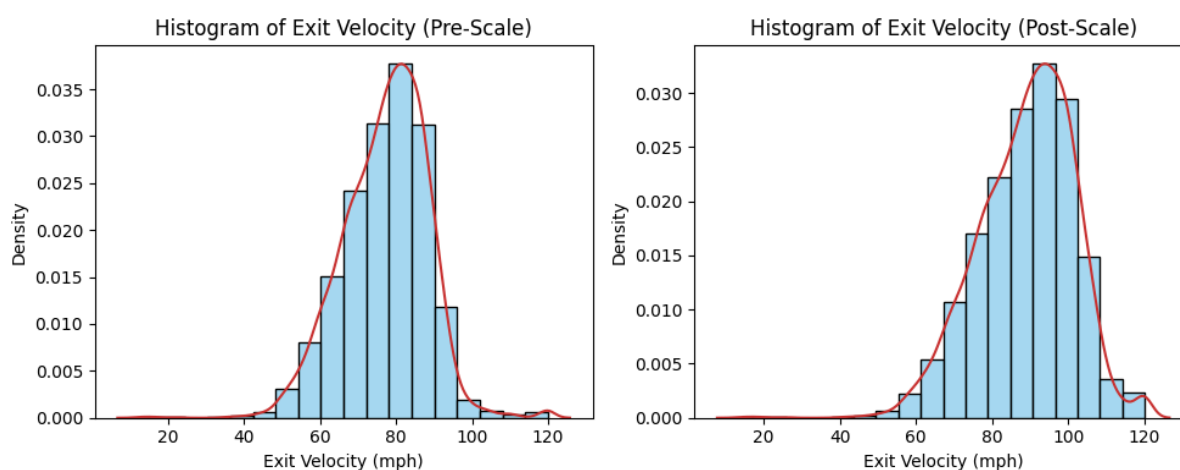


Figure 19: Side-by-side histograms of exit velocity (in mph) pre- and post-scale, overlaid with a KDE plot. The distribution (pre-scale) is centered a little lower than expected, around 78, but is scaled up to 89.

Credit to Tom Kim in the Discord for the idea of using the 0.4 to 0.6 second range and Dr. Meredith Wills for her extensive time and effort in helping me figure out the calculation for initial velocity.

³⁹ (*What is Exit Velocity?* – Baseball Connect, 2020).

Technical Detail 3: Opportunity + Hang Time

Hang time is the time difference between the landing point and the moment of contact. Landing point is the point at which the ball bounces for the first time, is caught, is deflected, or hits the wall. Whereas, opportunity time is the time difference between when the ball reaches its landing point and the moment the ball is *thrown* by the pitcher. Statcast uses opportunity time instead of hang time in their catch probability model in order to credit moments in which a fielder adjusts his position based on where the catcher sets up or what type of pitch is called⁴⁰.

Technical Detail 4: Distance Traveled + Needed

Distance traveled is calculated for the outfielder of interest from the moment of contact until the landing point ([Technical Detail 3](#)). It is calculated by summing the stepwise distances between consecutive timestamps to capture the actual, non-linear route taken. Distance needed is calculated as the straight-line distance from the fielder's starting position coordinates to the coordinates of the landing point, meaning it is the shortest distance the fielder has to go in order to catch the ball. Statcast uses distance needed instead of distance traveled in order to prevent fielders receiving any additional credit for taking longer-than-necessary routes to catch the ball¹¹.

Technical Detail 5: Direction

Direction is the angle between the fielder's starting position and the ball's landing point ([Technical Detail 3](#)). The angle is calculated in a similar fashion to launch angle ([Technical Detail 1](#)), as in, it is calculated using $\tan^{-1}$ in order to find Θ . However, there is an important distinction as to how direction works; since a fielder faces home plate, their forward-facing direction should be aligned with 270° on the unit circle, not 0° (this would imply that the fielder

⁴⁰ (Catch Probability | Glossary | MLB.com, 2025).

is facing to their right). As a result, 90° would represent directly behind them. Following the approach used in Statcast's Directional OAA, I split the directional angles into six different categories: In, In Left, In Right, Back, Back Left, Back Right⁴¹. Direction can be visualized in Figure 20 below along with its respective catch rate on BIPs.

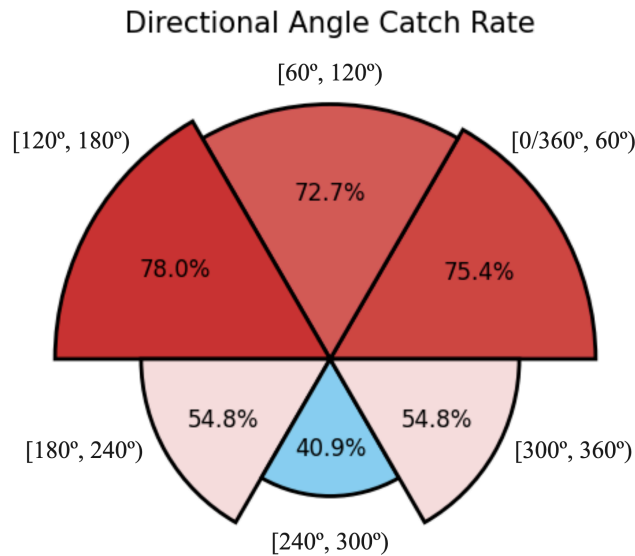


Figure 20: Visualizes direction with that direction's respective catch%. The darker the red and the more extended the slice, the greater the value is above 50%; the bluer and smaller the slice, the further it is below 50%. Visualizes the range of real unit circle angles from Table 6.

The specific angles to map each category can be seen in Table 6 below:

Direction	Real Unit Circle Range
Back Left	$[120^\circ, 180^\circ]$
Back	$[60^\circ, 120^\circ]$
Back Right	$[0/360^\circ, 60^\circ]$
In Left	$[180^\circ, 240^\circ]$
In	$[240^\circ, 300^\circ]$
In Right	$[300^\circ, 360^\circ]$

Table 6: Table of the translation of direction to its corresponding range of angles on the real Unit Circle.

⁴¹ (Statcast Directional Outs Above Average baseballsavant.com, 2025)

Direction is used in Statcast’s catch probability model as an adjustment based on the degree of difficulty, as “going back on the ball” tends to be harder for fielders⁴². However, I used it as a feature in my catch probability model ([Appendix B, Model 1](#)) instead. Plays where fielders move “back on the ball” do not appear especially difficult based on catch percentages, likely because the “Back” category spans a wide range of angles (60°–120° on the unit circle), including routine plays. Whereas, “In” shows lower catch rates, likely because these balls are hit in front of the fielder with shorter hang times, making them harder to judge.

Technical Detail 6: Sprint Speed

Sprint speed was calculated based on how Statcast calculates their sprint speed⁴³, by finding the greatest distance traveled in a one second window ($r = \frac{d}{t}$; $t=1 \text{ sec} \Rightarrow r = d$). This was done by looping through every iteration in a play and finding the distance traveled after one second⁴⁴.

Technical Detail 7: Wall Proximity

Wall proximity was calculated as the straight line distance from the ball’s landing point ([Technical Detail 3](#)) to the closest point on the wall. Since the home team’s stadiums were anonymized, I came up with my own quadratic polynomials to replicate the outfield walls. E.g., for team QEA’s stadium walls, I came up with the following piecewise function as an estimate of the wall: $w(x) = \{(x, -(\frac{x^2}{310}) + 400) \mid x < 0\} \cup \{(x, -(\frac{x^2}{395}) + 400) \mid x \geq 0\}$.

⁴² (Catch Probability | Glossary | MLB.com, 2025).

⁴³ (Sprint Speed (SS) | Glossary, n.d.).

⁴⁴ E.g., in a 20 fps game, a player’s sprint speed is the maximum distance out of the distances traveled from iteration=0 to iteration=20, iteration=1 to iteration=21,

Figure 21 below visualizes each team's approximate outfield walls.

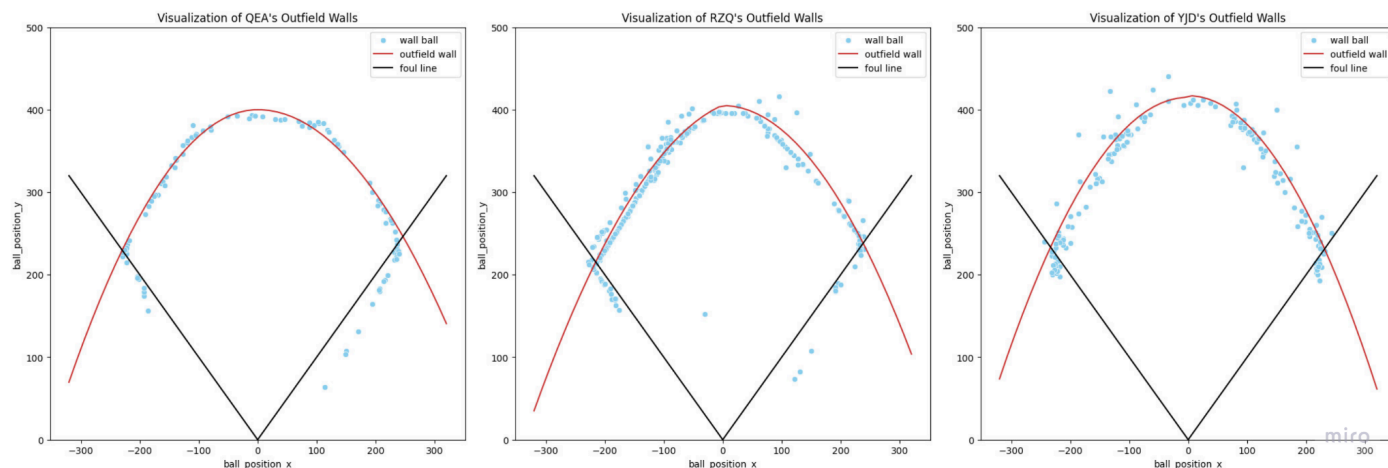


Figure 21: Each team's estimated stadium walls visualized. The left-side image is QEA's wall, the middle image is RZQ's walls, and the right-side image is YJD's walls. The red line represents the outfield wall, the black lines represent the foul lines, and the blue dots represent the actual balls that hit walls or went foul in each stadium.

With these “walls,” I found the closest point on the wall to the landing point ([Technical Detail 3](#)) by applying numerical optimization to find the x-coordinate on the wall that minimizes the distance to the ball's location. I then found the corresponding y-coordinate by inputting that x-coordinate into the estimated wall polynomial, and carried the calculation mentioned previously. Wall proximity is used in Statcast's catch probability model as a form of adjustment to account for difficult plays made near or at the wall, specifically within 8 feet horizontally and 8 feet vertically of the wall⁴⁵. In the same fashion as direction ([Technical Detail 5](#)), I used wall proximity as a feature in my catch probability model ([Appendix B, Model 1](#)).

Technical Detail 8: Jump

Jump is defined by Statcast as feet covered in the *correct* direction in the first three seconds after the pitcher releases the ball, and it is made up of three separate components: Reaction, Burst, and

⁴⁵ (Catch Probability | Glossary | MLB.com, 2025).

Route. Reaction is feet covered in *any* direction during the first 1.5 seconds⁴⁶. Burst is feet covered in *any* direction in the following 1.5 seconds to 3 seconds⁴⁶. And Route is feet covered against the direct route in the first three seconds⁴⁶. Each of those components are then compared to an average, which were computed with models ([Appendix B, Model 2](#)). Reaction and Burst were calculated finding the total stepwise distance on a single route, and the calculation of Route can be seen in Figures 22, 23, and 24.

$$x = \frac{(Direct\ Route\ Distance)^2 + (Straight\ Line\ Distance\ Traveled_{0\ to\ 3\ sec})^2 - (Distance\ Needed_{3\ sec\ to\ landing\ pt})^2}{2 * Direct\ Route\ Distance}$$

Figure 22: The formula used to calculate Route.

The formula is based on the cosine ratio for right triangles and the law of cosines. Figure 23 below is a simplified version of the derivation:

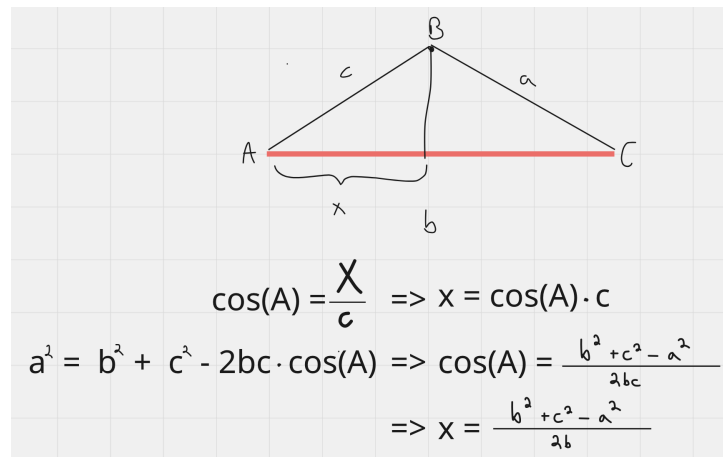


Figure 23: Visualizes the math behind calculating Route with arbitrary variables, using the law of cosines.

Figure 24 below is what can be seen in Figure 23 above and the actual application of the derivation above.

⁴⁶ (Andrews, 2022).

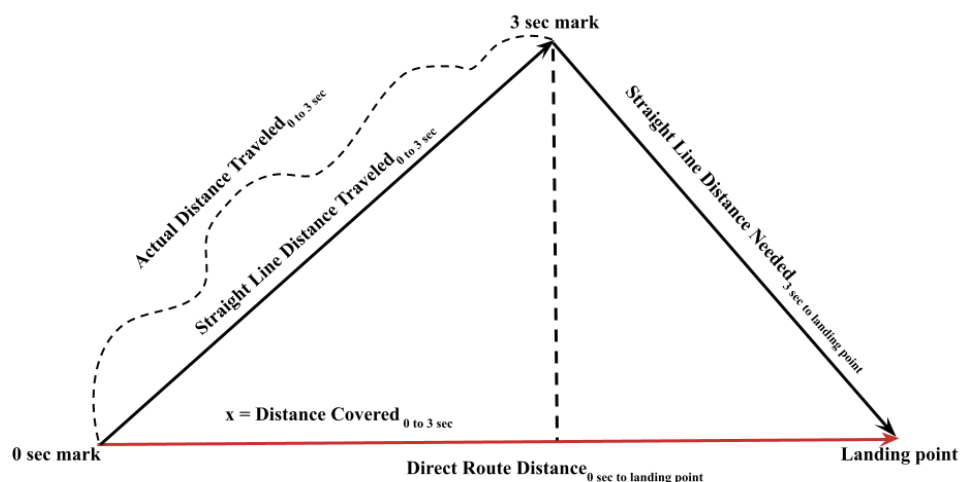


Figure 24: Visualizes the route component in jump. Shows the vectors of the outfielder's route based on the 0 sec mark, 3 sec mark and landing point, including the projection of the 3 sec mark to the direct route.

In the event that the landing point *does* occur less than or equal to three seconds after the ball is hit, Route is the direct distance between the fielder's starting position and their position the moment the ball lands.

Technical Detail 9: Data Filtering

I filtered the dataset to only include plays where an outfielder first touched the ball after it was put in play, considering only the timeframe from the pitch to when the outfielder acquired the ball. Plays involving errors, infield touches before the outfield, or subsequent touches (e.g., quick tosses between fielders) were excluded, keeping only the first outfielder touch. Ground balls that reached the outfield were also removed. Ground balls were classified as balls with launch angles ([Technical Detail 1](#)) below 10° that were not caught for an out⁴⁷. Balls with hang times ([Technical Detail 3](#)) under 1.5 seconds were removed since these are virtually uncatchable⁴⁸. And finally, I filtered out wall-ball plays ([Technical Detail 7](#)) with a z-coordinate above 10 ft, as

⁴⁷ (Launch Angle (LA) | Glossary, n.d.).

⁴⁸ (Zimmerman, 2014).

catches within the data set never exceeded 8.5 ft in height when within 8 ft of the wall. I also cleaned the `game_info` table to account for missing plays within games ([Appendix D](#)). E.g., if plays 3–6 were missing between plays 1, 2, and 7, I imputed the missing plays back into the dataset and filled in the appropriate IDs. This ensured that every play from every game could be properly attributed to a player.

Appendix D: Description of the Data

Below is a more detailed description of the data that I used. Note: these are not all of the tables and variables available to us but ones I used extensively.

game_events: this table contains an id string for each game, an id for a play that occurs during that game, a timestamp every 50 ms or 33 ms (depending on the camera's fps that game), and encoding for the player's position and the event code that occurred during that play.

ball_pos: this table contains 3D tracking data for the ball in the x-y-z plane from the back of home plate. A positive x-value indicates the ball is towards first base, a positive y-value indicates the ball is towards second base, and a positive z-value indicates the ball is above the z=0 plane (this does not necessarily mean the ground). All coordinates are measured in feet.

player_pos: this table contains 2D tracking data for each player in the x-y plane from the back of home plate. The x-values and y-values follow the same description as the x- and y-values in *ball_pos*. Also measured in feet.

game_info: this table contains information about the game itself. The information it contains are things like the IDs of the players, what inning it is, what play of the game it is, etc.